Newton’s First Law and Newton’s Second Law

1. In each of the following cases say whether the forces acting on the object are in equilibrium by deciding whether its velocity is changing.
   (a) A car that has been stationary, as it moves away from a set of traffic lights.
   (b) A parachutist descending at a constant rate.
   (c) A rock falling freely.
   (d) A book resting on a table.
   (e) A car going round a corner.

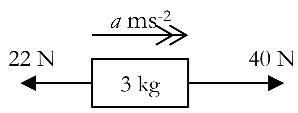
2. What is the acceleration caused by a 30 N force on a 1.5 kg book?

3. What resultant force is required to accelerate:
   (a) a 3 kg sandbag at 2 ms^{-2},
   (b) a 40 g beanbag at 5 ms^{-2},
   (c) a 2 tonne car at 3.5 ms^{-2},
   (d) a jumping flea of mass 0.05 mg at 1750 m s^{-2}.

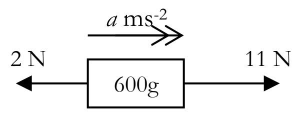
4. A 18 N force on a shot put causes an acceleration of 4.5 m s^{-2}. Find the mass of the shot put.

5. Find the missing forces and accelerations:

(a)



(b)



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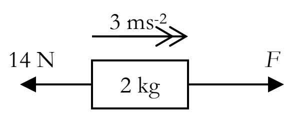
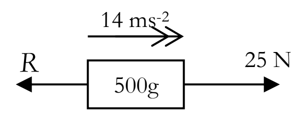
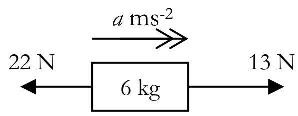


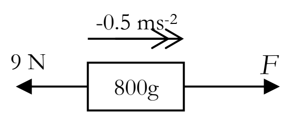
(d)



(e)



(f)



6. A man pulls a block of mass 4 kg with force *F* N against a resistance force of 20 N. The block accelerates at 2.5 ms^{-2}.
   (a) Draw a force diagram
   (b) Hence find *F*.

7. A man pulls a block of mass 750g with force 21 N against a resistance force of 15 N. Find the acceleration.
   (a) Draw a force diagram
   (b) Hence find the acceleration *a*.

8. A man pulls a block of mass *m* kg with force 70 N against a resistance force of 19 N. The block accelerates at 3 ms^{-2}. Find *m*.
   (a) Draw a force diagram
   (b) Hence find *m*.

9. A man pulls a block of mass 18 kg with force 100 N against a resistance force of *R* N. The block accelerates at 1.5 ms^{-2}.
   (a) Draw a force diagram
   (b) Hence find *R*.

10. A man pulls a block of mass 8 kg with force *F* against a resistance force of 50 N. The block decelerates at 1.5 ms^{-2}.
   (a) Draw a force diagram
   (b) Hence find *F*.

11. The engine of a car of mass 1.6 tonnes can produce a driving force of 2000 N.
    (a) Ignoring any resistance forces, find the car's resulting acceleration.
    (b) In fact the car's acceleration is 1 ms^{-2}; find the resistance force.

12. A police car is chasing a stolen car. Both cars have the same mass. The police car produces a driving force of 1800N and is acted on by a resistance force of 300N. The stolen car produces a driving force of 1900N and is acted upon by a resistance force of 200 N. Choose which of the following statements is true:
    (a) The police car will eventually catch up the stolen car;
    (b) The stolen car will get away from the police car;
    (c) More information is needed.

    Another police car is chasing a different stolen car. Again, both cars have the same mass. This police car produces a driving force of 2000N and is acted on by a resistance force of 300N. This stolen car produces a driving force of 1800N and is acted upon by a resistance force of 200 N. Choose which of the following statements is true:
    d) The police car will eventually catch up the stolen car;
    e) The stolen car will get away from the police car;
    f) More information is needed.

13. A woman of mass 65 kg stands in an accelerating lift. Draw a diagram showing the forces acting *on the woman*, and find the force that the floor of the lift exerts on her in the following cases:
    (a) The lift remains at rest;
    (b) The lift is moving upwards at a constant speed of 2 ms^{-1};
    (c) The lift is accelerating upwards at 0.5 ms^{-2};
    (d) The lift is accelerating downwards at 1 ms^{-2}.

14. Later, the woman returns to the lift, this time carrying a box of mass 2 kg. Draw a diagram showing the forces acting *on the box*. Find the force that *the woman exerts on the box* in the following cases:
    (a) The lift is moving upwards at a constant speed of 2 ms^{-1};
    (b) The lift is accelerating upwards at 0.5 ms^{-2};
    (c) The lift is accelerating downwards at 1 ms^{-2}.

    Also draw a diagram showing the forces acting *on the woman*. [Hint: it's **not** the box's weight!] Find the force that *the box exerts on the woman* in the following cases:
    (d) The lift is moving upwards at a constant speed of 2 ms^{-1};
    (e) The lift is accelerating upwards at 0.5 ms^{-2};
    (f) The lift is accelerating downwards at 1 ms^{-2}.

    What do you notice?